

Represent Rational Numbers on the Coordinate Plane

Lesson 7-5

Name: Type your name **Date:** Today's date **Class:** Class period

LOST COVE MISSION

Captain Vega's Coordinates

You are the navigator on Captain Vega's expedition. Instead of 'X marks the spot,' she hid the treasure route as ordered pairs scrawled across the first quadrant of her chart, using only positive coordinates. Plot every point in the right order and the lost cove treasure is yours.

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can plot and identify points on the coordinate plane using ordered pairs.

Language Objective: I can explain my work using the words coordinate plane, ordered pair, origin, and quadrant.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Graph on the Coordinate Plane

Coordinate plane

Ordered pair

Origin

Quadrant

Reflection

Integer



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— Plot a point by moving horizontally to its x-coordinate and vertically to its y-coordinate.

2

The grid made by a horizontal and a vertical number line is the

.

3

Two numbers (x, y) that tell the location of a point are an

.

4

The point (0, 0) where the x-axis and y-axis cross is the

.

5 One of the four regions of the coordinate plane is a .

6 A flip of a point or shape over an axis is a .

7 A whole number or its opposite, with no fraction part, is an .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How do ordered pairs help the drone navigate? What shape does the flight path make?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Graph on the Coordinate Plane** — Plot a point by moving horizontally to its x-coordinate and vertically to its y-coordinate.

Representar en el plano de coordenadas — Ubicar un punto moviéndose horizontalmente hasta su coordenada x y verticalmente hasta su coordenada y .

- ☐ **Coordinate plane** — A grid with a line going across and a line going up to plot points.

Plano cartesiano — Una cuadrícula con una línea horizontal y una vertical para marcar puntos.

- ☐ **Ordered pair** — Two numbers (x, y) that tell where a point is on a grid.

Par ordenado — Dos números (x, y) que dicen dónde está un punto en una cuadrícula.

- ☐ **Origin** — The point $(0, 0)$ where the two grid lines cross.

Origen — El punto $(0, 0)$ donde se cruzan las dos líneas de la cuadrícula.

- ☐ **Quadrant** — One of the four parts of a coordinate grid, numbered I to IV.

Cuadrante — Una de las cuatro partes de una cuadrícula de coordenadas, numeradas I a IV.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The drone uses ordered pairs because the first number controls ____ movement and the second controls _____. The path makes a ____ shape.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: We start at the origin and move along the x-axis first because ____.

Evidence: Student states the x-coordinate is the sideways (horizontal) move and the y-coordinate is the up/down (vertical) move, both starting from the origin.

Reasoning: This evidence connects the definition of coordinate plane to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The drone uses ordered pairs because the first number controls ____ movement and the second controls _____. The path makes a _____ shape.
- My answer is _____.

Mi respuesta es _____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is _____.
- I know because _____.

Mi afirmación es _____.

Lo sé porque _____.

- So _____.

Entonces _____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.